

PROGRAM FOR OPERATORS

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PYTHON OPERATORS - ALL TYPES

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1. ARITHMETIC OPERATORS

print("==== 1. ARITHMETIC OPERATORS ====")

a = 10

b = 3

print("Addition :", a + b) # +

print("Subtraction :", a - b) # -

print("Multiplication :", a * b) # *

print("Division :", a / b) # /

print("Floor Division :", a // b) # //

print("Modulus :", a % b) # %

print("Exponent :", a ** b) # **

2. ASSIGNMENT OPERATORS

print("\n==== 2. ASSIGNMENT OPERATORS ====")

x = 10

print("x =", x)

x += 5

print("x += 5 :", x)

x -= 3

print("x -= 3 :", x)

x *= 2

print("x *= 2 :", x)

x /= 4

print("x /= 4 :", x)

x //= 2

print("x //= 2 :", x)

x %= 3

print("x %= 3 :", x)

x **= 2

print("x **= 2 :", x)

3. COMPARISON OPERATORS

print("\n===== 3. COMPARISON OPERATORS =====")

p = 10

q = 20

print("p == q :", p == q) # Equal

print("p != q :", p != q) # Not equal

print("p > q :", p > q) # Greater than

print("p < q :", p < q) # Less than

print("p >= q :", p >= q) # Greater than or equal

print("p <= q :", p <= q) # Less than or equal

4. LOGICAL OPERATORS

print("\n===== 4. LOGICAL OPERATORS =====")

age = 20

has_id = True

print("AND :", age >= 18 and has_id)

print("OR :", age >= 18 or has_id)

print("NOT :", not has_id)

5. IDENTITY OPERATORS

print("\n===== 5. IDENTITY OPERATORS =====")

list1 = [1, 2, 3]

list2 = list1

list3 = [1, 2, 3]

print("list1 is list2 :", list1 is list2)

print("list1 is list3 :", list1 is list3)

print("list1 is not list3 :", list1 is not list3)

6. MEMBERSHIP OPERATORS

print("\n===== 6. MEMBERSHIP OPERATORS =====")

numbers = [10, 20, 30, 40, 50]

print("20 in numbers :", 20 in numbers)

print("60 in numbers :", 60 in numbers)

print("60 not in numbers :", 60 not in numbers)

OUTPUT:

===== 1. ARITHMETIC OPERATORS =====

Addition : 13

Subtraction : 7

Multiplication : 30

Division : 3.3333333333333335

Floor Division : 3

Modulus : 1

Exponent : 1000

===== 2. ASSIGNMENT OPERATORS =====

x = 10

x += 5 : 15

x -= 3 : 12

x *= 2 : 24

x /= 4 : 6.0

x //= 2 : 3.0

x %= 3 : 0.0

x **= 2 : 0.0

===== 3. COMPARISON OPERATORS =====

p == q : False

p != q : True

p > q : False

p < q : True

p >= q : False

p <= q : True

===== 4. LOGICAL OPERATORS =====

AND : True

OR : True

NOT : False

===== 5. IDENTITY OPERATORS =====

list1 is list2 : True

list1 is list3 : False

list1 is not list3 : True

===== 6. MEMBERSHIP OPERATORS =====

20 in numbers : True

60 in numbers : False

60 not in numbers : True